Wet-wall cyclone bioaerosol sampling and optical detection instrument

Equipment technical requirements (complete instrument)

Wet-wall cyclone bioaerosol sampling and optical detection instrument — English version of the complete-instrument requirements.

This document specifies the equipment requirements for the complete instrument of the Echo Yeast project, in eight parts: scope, normative references, terms and definitions, technical requirements, test methods, inspection rules, marking and packaging, and notes. The complete instrument is the overall goal of the project; the temperature-control module, the optical plug-in modules and the signal-acquisition circuit built so far follow the interfaces and figures set out here, and the display and host-computer functions will be implemented to this document once the core functions have been verified.

1 Scope

This specification defines the terms and definitions, technical requirements, test methods and inspection rules of the wet-wall cyclone bioaerosol sampling and optical detection instrument, together with its marking, packaging, transport and storage. The instrument samples virus aerosols from ambient air, collects them in liquid, performs yeast whole-cell sensing and reads out the optical signal. It is intended for laboratory use and for use as a plug-in module of an air disinfection unit.

2 Normative references

The following documents are indispensable for the application of this specification. For dated references, only the dated edition applies.

GB/T 38517-2020 Particles — General requirements for bioaerosol sampling and analysis

GB/T 1804-m General tolerances — Tolerances for linear and angular dimensions without individual tolerance indications

GB 9706.1 Medical electrical equipment — Part 1: General requirements for safety

WS 648-2019 General hygiene requirements for air disinfection units (if used as a plug-in module of such a unit)

ISO 10993 Biological evaluation of medical devices

USP Class VI biocompatibility requirements

3 Terms and definitions

The terms and definitions given in GB/T 38517-2020 apply to this specification.

4 Technical requirements

4.1 Overall performance

Item	Requirement
Sampling method	Centrifugal (cyclone) liquid sampling, in accordance with clause 4.2.4 of GB/T 38517-2020
Sampling flow rate	100–300 L/min, adjustable, matched to the cyclone sampler
Sampling time	15–30 min for virus aerosols, in accordance with clause 4.6.4.6 of GB/T 38517-2020
Working volume	20 mL of liquid medium containing yeast, pre-loaded into the collection cup before sampling
Target analyte	HA protein on the surface of influenza A virus (H1N1)
Detection principle	After the HA protein binds the receptor displayed on the yeast surface, receptor auto-inhibition is relieved and the FUS1 promoter is activated through the MAPK pathway, inducing fluorescent protein expression; 485 nm serves only as the readout excitation wavelength, and the 528 nm fluorescence intensity gives a qualitative result
Detection type	Qualitative (influenza A virus present / absent)
Response time	Under ideal laboratory conditions a 528 nm fluorescence signal is detectable 5–15 minutes after the sample is added; the response time of the complete instrument is to be verified experimentally
Operating temperature	30 ± 0.5 °C (yeast culture temperature)
Power supply	AC 220 V ± 10%, 50 Hz; or DC 24 V / 5 A
Dimensions	Approx. 300 × 250 × 200 mm (subject to the final structural design)
Weight	≤ 5 kg

4.2 Structural requirements

4.2.1 Enclosure (3D printed)

The enclosure is manufactured by 3D printing, with the following requirements:

Item	Requirement
Printing material	PETG, ABS or flame-retardant PC-ABS; if photopolymer resin is used it must be of a biocompatible grade that does not deform or yellow in long-term use
Printing process	FDM or SLA, layer height 0.1–0.2 mm, wall thickness ≥ 2 mm, infill ≥ 20%
Dimensional accuracy	Critical mounting holes ±0.2 mm, general external form ±0.5 mm
Surface finish	External surfaces sanded smooth; internal surfaces painted matt black (or lined with light-absorbing flock) to reduce stray light
Light tightness	The optical detection region must be completely light-tight; every seam and vent needs a labyrinth structure or light-blocking foam so that no ambient light enters
Heat dissipation	Vents are provided but must remain light-tight; a labyrinth duct or a light-blocking mesh may be used
Mounting interfaces	M3/M4 heat-set inserts or threaded holes for the internal modules, screen, switches and so on
Flame retardancy	If the instrument is to pass GB 9706.1, the enclosure material must meet the corresponding flame-retardant rating (for example UL94 V-0)
Other	The enclosure should be easy to open, with access for debugging, a USB port and a power socket

4.2.2 Inlet vessel (combined cyclone collection and optical detection chamber)

The inlet vessel is a custom injection-moulded part in medical optical-grade COC, with the following structural requirements:

Item	Requirement
Material	Medical optical-grade COC (TOPAS 6017 or equivalent), transmittance ≥ 90% at 485/528 nm, very low intrinsic fluorescence
Thread	M28×1.5-6g, checked with go/no-go gauges; must screw onto the sampler smoothly and seal
Overall height	103 mm

Item	Requirement
Cylindrical section	Inner Ø32 mm, outer Ø36 mm, height 33 mm
Conical section	Vertical height 54.21 mm, half-angle 12° (included angle 168°), upper inner Ø32 mm, lower inner Ø12 mm
Base	Inner radius R6 mm, outer skirt R8 mm, outer base Ø36 mm
Working liquid level	The 20 mL level is about 50 mm from the cone apex and must fully cover the optical windows
Optical windows	Two, at 90° to each other, Ø10 mm circular flat faces, centres 40 mm from the cone apex, each face inclined 12° to the horizontal (normal to the cone surface)
Window wall thickness	2 mm; inner and outer faces are parallel planes, parallelism ≤ 0.02 mm
Window surface	Ra ≤ 0.05 μm, flatness ≤ λ/4 at 528 nm, edge chamfer C0.3
Window position	Must be fully immersed below the 20 mL working liquid level
Sealing	Once connected to the cyclone sampler, pressurised to 0.05 MPa and held for 30 s with no bubble leakage
Cleanliness	Free of oil, particles and mould-release residue before dispatch; optical faces covered with protective film
Sterilisation	Withstands ethylene oxide or gamma sterilisation; if autoclaving is required, freedom from deformation must be verified in advance

4.3 Temperature control module

Item	Requirement
Heating method	PI heating film, 100 × 100 mm, 24 V / 50–100 W, bonded to the underside of the aluminium block
Heater block	High-purity aluminium 6061 or 7075, CNC machined, bore Ø22–25 mm, depth 30 mm, 4–9 bores
Temperature sensor	PT100 platinum RTD, class A, 3-wire, embedded in a central blind bore in the aluminium block and filled with thermal grease

Item	Requirement
Temperature transmitter	MAX31865 module, SPI interface, 16-bit ADC
Control range	Room temperature to 40 °C, set point 30 °C
Temperature accuracy	±0.5 °C
Over-temperature protection	KSD9700 thermal cut-off, normally open, trip temperature 60 °C, 250 V / 10 A, cuts the heater at hardware level
Power drive	IRF520 or AO3400 MOSFET, PWM power control under PID
Heat dissipation	40 mm or 50 mm quiet fan, 12 V; aluminium heat sink matched to the fan size
Thermal interface	Heat-resistant thermal silicone pad, 5 mm thick; thermal grease ≥ 3 W/(m·K)

4.4 Optical detection module

Item	Requirement
Excitation source	485 nm high-brightness LED, 5 mm or 10 mm, constant-current driven, 350 / 500 mA adjustable
Collimation	A collimating lens in front of the LED produces a parallel beam, spot Ø5–8 mm, divergence ≤ 5°
Excitation filter	480/30 band-pass, removes LED stray light
Emission filter	530/20 band-pass, OD ≥ 6 at 485 nm, transmittance ≥ 90% at 528 nm
Photodetector	BPW34 or SFH203 silicon photodiode, responsive at 528 nm
Transimpedance amplifier	OPA350 or LM358 operational amplifier in a transimpedance configuration
Optical layout	90° geometry; the excitation beam and the fluorescence share the two optical windows
Light shielding	Black heat-shrink tubing or a 3D-printed shroud to exclude ambient light
Optical signal acquisition	ADS1115 module, 16-bit ADC, 4 differential channels, I^2C interface

Item	Requirement
Limit of detection	To be verified experimentally; target ng/mL or PFU/mL level
Specificity	Specific response to the HA protein of the target virus, to be verified against negative controls

4.5 Main control and display module

Item	Requirement
Main control board	STM32F103C8T6 core board (recommended) or an ESP32 development board
Display	4.3-inch or 7-inch TFT capacitive touch screen over a serial link (DWIN / Nextion)
User input	KY-040 rotary encoder with push button, for setting the temperature and other parameters
Alarm	Active buzzer, 5 V, sounding on over-temperature or an optical fault
Programming interface	USB-to-TTL module, CH340 or CP2102
Data storage	Real-time display of the temperature and fluorescence intensity curves; data export supported

4.6 Communications and host computer

Item	Requirement
Wired communications	USB-to-TTL, CH340 or CP2102, 5 V / 3.3 V dual voltage
Wireless communications (optional)	HC-05 Bluetooth module or ESP8266 Wi-Fi module
Host software	Python + PyQt5 + PySerial, open source and free
Functions	Real-time display of temperature and fluorescence intensity curves; export to Excel
Protocol	Custom serial protocol covering temperature setting, optical signal readout and status queries

4.7 Power supply and auxiliary components

Item	Requirement
Switching supply	24 V / 5 A (120 W) with an ON/OFF switch
Buck module	LM2596, 24 V to adjustable 5 V / 3.3 V
Circuit construction	830-hole breadboard or universal protoboard
Wiring	Dupont jumpers, male–male, male–female and female–female
Filtering	Electrolytic capacitors 100 μF / 16 V, 10 μF / 50 V
Resistors	Common values such as 1 kΩ, 10 kΩ and 100 Ω
Indicators	5 mm red / green LEDs for power and run status
Terminal blocks	2-way / 3-way PCB terminal blocks

4.8 Safety and biosafety

Electrical safety: complies with the general safety requirements of GB 9706.1 for medical electrical equipment.

Over-temperature protection: dual protection by a hardware-level KSD9700 thermal cut-off and a software alarm.

Biosafety: work involving live virus must be carried out in a laboratory of the appropriate biosafety level, with operators wearing the corresponding protective equipment, in accordance with clause 4.5.4 of GB/T 38517-2020.

Sample handling: after sampling, liquid samples are stored and transported as required by clause 4.6.7 of GB/T 38517-2020 (0–10 °C, delivered for testing within 10 h).

Disinfection and sterilisation: the sampling head, the vessel and the operators' hands are disinfected before and after sampling, in accordance with clause 4.5.5 of GB/T 38517-2020.

Exhaust air: the exhaust must pass through HEPA H13 or higher-grade filtration to prevent aerosol leakage.

4.9 Software functions

PID temperature control: the PWM output to the heating film is regulated to hold 30 ± 0.5 °C.

Multi-channel optical acquisition: the 485 nm LED is pulsed on a timer, the photodiode voltage is read and converted into fluorescence intensity.

Serial protocol: the host computer can send the temperature set point and sampling time and read data back.

Data display: a TFT screen shows the temperature, the fluorescence intensity curve and the countdown in real time.

Data export: the host computer can export the data in Excel format.

Alarms: a buzzer sounds and an error code is displayed on over-temperature, a blocked optical path or a communication fault.

5 Test methods

Item	Test method
Temperature accuracy	Set 30 °C; once stable, measure the block temperature with a calibrated thermometer; deviation ≤ ±0.5 °C
Window transmittance	Measure transmittance at 485 nm and 528 nm with a spectrophotometer; ≥ 90%
Window background fluorescence	Illuminate blank medium with the 485 nm LED and detect through the 528 nm filter; background signal ≤ 3 SD of the negative control

Item	Test method
Airtightness	Pressurised to 0.05 MPa, held for 30 s, no bubble leakage
Thread gauges	Checked with M28×1.5-6g go/no-go gauges
Sampling flow rate	Calibrated with a flow meter; error ≤ ±5%
Detection sensitivity	Test with known concentrations of HA protein or virus samples to establish the limit of detection
Biocompatibility	Tested to ISO 10993 or USP Class VI

6 Inspection rules

First-article inspection: the first article of each batch requires dimensional, transmittance, airtightness and biocompatibility test reports.

Factory inspection: every unit is checked for temperature accuracy, optical background, airtightness and thread gauge fit.

Type inspection: the full set of tests is carried out when a new product is finalised or the design is changed.

7 Marking, packaging, transport and storage

Marking: the nameplate gives the model, name, power supply, manufacturer and date of manufacture.

Packaging: optical surfaces are covered with protective film and packed separately, protected against dust, scratches and static; the certificate of conformity, material certificates and inspection reports are enclosed.

Transport: avoid high temperature, moisture and severe vibration.

Storage: ambient temperature 0–40 °C, relative humidity ≤ 80%, no corrosive gases.

8 Notes

The instrument is used either as a plug-in module for a cyclone bioaerosol sampler or as a stand-alone device.

If used as a plug-in module of an air disinfection unit, it must additionally meet WS 648-2019 and match the airflow of the host unit.

All unspecified tolerances per GB/T 1804-m; unspecified fillets R0.5, unspecified chamfers C0.5.

The inlet vessel has two optical windows set at 90° to each other, shown in the rotated section A-A; in manufacture the two windows are 90° apart.

The enclosure is 3D printed; the material must meet the requirements for light tightness, heat dissipation, flame retardancy and structural strength.

The firmware and hardware versions of the instrument must be compatible with the host computer; the protocol is given in the Communication Protocol Manual.

The “working volume”, “target analyte” and “response time” clauses of this specification reflect the latest experimental data: the working volume is 20 mL of liquid medium containing yeast, pre-loaded into the collection cup before sampling; the target analyte is the HA protein on the surface of influenza A virus (H1N1); and under ideal laboratory conditions a 528 nm fluorescence signal is detectable 5–15 minutes after the sample is added, while the response time of the complete instrument is to be verified experimentally.